

FINC409/609

Fixed Income Securities

Prof. Matt Fleckenstein

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Office Hours: Tue, 5.00pm - 6.00pm, 310 Purnell Hall or by appointment.

COURSE DESCRIPTION

This course provides an introduction to fixed income markets and the analytical tools used by market participants. Fixed income markets have evolved significantly in recent years, particularly following the global financial crisis. Regulatory changes, unconventional monetary policy and extended periods of very low or near-zero nominal interest rates, increased government borrowing, shifts in the empirical behavior of U.S. Treasury securities, and the introduction of new fixed income instruments have all influenced how these markets function and how fixed income securities are priced and risk-managed.

Fixed income is a broad topic that cannot be covered comprehensively in a single semester. Accordingly, the course is selective and focuses on the core concepts and methods that underpin fixed income valuation and risk management. With a strong foundation in the underlying theory and market conventions, students will be well-positioned to learn additional institutional details and specialized products in professional settings.

The course is quantitative and emphasizes practical implementation. Students will work extensively with formulas, numerical methods, and real-world valuation and hedging techniques widely used in industry, including by banks and asset managers.

LEARNING OBJECTIVES

After completing this course, students will be able to:

- Describe the structure of bond markets, the main fixed income instruments, and the conventions used to quote, trade, and price fixed income securities.
- Explain the U.S. Treasury market, including major Treasury instruments and the mechanisms through which Treasuries are issued and traded.
- Apply time-value-of-money principles to price fixed income cash flows and interpret discount factors and their role in valuation.
- Use replicating portfolios and the law of one price to value fixed income securities and identify arbitrage relationships.
- Analyze the term structure of interest rates and use it to price bonds and other fixed income instruments.

- Explain the bond price-yield relationship and quantify interest rate sensitivity using standard measures.
- Compute and apply duration and convexity to measure interest rate risk and to hedge single-bond positions and bond portfolios.
- Implement and evaluate common fixed income trading strategies.
- Understand the mechanics and uses of forward and futures contracts in fixed income markets.
- Price and interpret key interest rate derivatives, especially interest rate swaps, and apply them to hedge interest rate risk.

CLASS TIMES

FINC 409/609	
Day(s)	Tue, Thu
Time	Section 010: 9.35am-10.55am Section 011: 11.10am-12.30pm
Location	Gore Hall 219

OFFICE HOURS

Section	FINC 409/609
Day(s)	Tuesdays
Time	5.00pm-6.00pm
Location	310 Purnell Hall

PREREQUISITES

FINC 311 (Principles of Finance) and 314 (Investments) are prerequisites. A good understanding of basic algebra, calculus, and statistics is key to mastering the material covered in this course. It will also be useful if you already have some familiarity with: Present Value, compounding of interest rates, the CAPM, Black-Scholes. If you are unfamiliar with these concepts, it will simply mean a little bit more work for you as we go through the associated topics in class.

COURSE WEBPAGE

<https://fleckenstein-m.github.io/FINC462-662-SP2026/>

COURSE MATERIALS

All course materials will be posted on the course webpage at <https://fleckenstein-m.github.io/FINC462-662-SP2026/>.

The recommended textbook is Bond Markets, Analysis, and Strategies, Tenth Edition by Frank J. Fabozzi and Francesco A. Fabozzi.

Additional resources are

- Fixed Income Markets and Their Derivatives, 3rd edition, Suresh Sundaresan.
- Fixed Income Securities: Tools for Today's Markets, 3rd edition, Bruce Tuckman and Angel Serrat.
- Fixed Income Securities: Valuation, Risk, and Risk Management, Pietro Veronesi.
- Fixed Income Analysis, 3rd edition, Barbara S. Petitt, Jerald E. Pinto, and Wendy L. Pirie.

GRADING POLICY

- Your final course grade is based on three components: (i) two midterm exams, (ii) a final exam, (iii) class participation.
- Grades are assigned *on a curve*. This means your overall performance will be *ranked relative to other students* in the section(s), using the weighted components below.

ASSESSMENT WEIGHTS

- 25% Midterm 1
- 25% Midterm 2
- 30% Final Exam
- 20% Class Participation
- Your weighted total determines your *percentile rank*, which is then converted into a *final letter grade*.

Grade	Percentile Rank
A	94.0 and above
A-	90.0 to <94.0
B+	87.0 to <90.0
B	84.0 to <87.0
B-	80.0 to <84.0
C+	77.0 to <80.0
C	74.0 to <77.0
C-	70.0 to <74.0

Grade	Percentile Rank
D+	67.0 to <70.0
D	64.0 to <67.0
D-	61.0 to <64.0
F	<61.0

EXAM SCHEDULE

1. Midterm 1

- Date: *March 12, 2026*
- Time: 9.35am-10.55am (Section 010); 11.10am-12.30pm (Section 011)
- Location: Gore Hall Hall 219

2. Midterm 2

- Date: *April 14, 2026*
- Time: 9.35am-10.55am (Section 010); 11.10am-12.30pm (Section 011)
- Location: Gore Hall Hall 219

3. Final Exam

- Section 010:
 - Date: *May 19, 2026*
 - Time: 8.00am-10.00am
 - Location: Gore Hall 219
- Section 011:
 - Date: *May 16, 2026*
 - Time: 11.30am-1.30pm
 - Location: Gore Hall 219

ASSESSMENT COMPONENT: CLASS PARTICIPATION

- *Attendance*: You are required to come to class and be prepared to answer questions on the material.
- *Homework assignments* will be given nearly every week and will be announced in class and posted on the course webpage. You are expected to work on the homework assignments in groups of three. Homework assignments must be submitted before the lecture starts on the date the assignment is due. In addition, I reserve the right to ask the groups to solve up to three questions from the homework assignments in front of the class. The groups can select the problems they would like to present.

- *Quizzes*: I also reserve the right to give *Quizzes* in class. The in-class quizzes will be modeled on the examples in the lecture material after the topic is covered.

DATA ASSIGNMENTS

There will be two data assignments during the semester. These two assignments are optional.

COURSE OUTLINE

1. Introduction to Fixed Income
2. U.S. Treasury Bond Market
3. Time Value of Money
4. Bond Valuation
5. Term Structure of Interest Rates
6. Duration & Convexity
7. Spot Rates, Forward Rates, Expectations Hypothesis
8. Forward Rate Agreements and Futures Contracts
9. Interest Rate Swaps & Floating Rate Bonds
10. Relative value trades between U.S. Treasury notes and Inflation Protected Securities (TIPS)

POLICIES

FAIRNESS

I will make every effort to treat students with dignity and fairness and to be sensitive to the diversity that exists within the student body. Students with disabilities who request help will be given reasonable accommodation with the assistance of the Office of Disability Support Services (<http://sites.udel.edu/dss>). Please feel free to speak to me in private about any related issues.

ACADEMIC MISCONDUCT

Students are expected to abide by the University of Delaware Code of Conduct. Among other things, please do not submit plagiarized work or give or receive information during exams, and please provide proper acknowledgment in work when applicable. *Academic misconduct includes the use of websites such as CourseHero and Chegg.* The University's General Counsel has provided the following regarding such websites.

Notice: The University of Delaware owns a non-exclusive right to faculty materials that will be given to you during the course. These materials include ideas and comments provided by the instructor during class; test

questions; handouts and course materials; and other materials representing the copyright-protected intellectual property of course instructors. You are prohibited from entering into relationships with web sites such as Course Hero, Notehall, or similar sites under which you agree, in exchange for the payment of a fee or salary, to post these materials online. Violations of this prohibition will be referred to the Student Conduct Office for investigation and disciplinary action under UD's Code of Conduct.

The use of ChatGPT (or other similar AI) to complete assignments, cases, or other parts of the course will be considered academic misconduct unless you personally wrote the underlying code to the AI.

CLASSROOM CONDUCT

Please silence all cell phones and do not text in class. Laptops and tablets will be permitted in class for note-taking purposes, but please do not distract your classmates by using your laptop to surf the internet or conduct other business during lectures. I reserve the right to ban the use of laptops and tablets if they become a distraction. I also ask that students remain professional and respectful during class discussions.

QUESTIONS OUTSIDE OF CLASS

For any questions that have not been sufficiently answered in class, please e-mail me or come to office hours.

ABSENCES FROM EXAMS

Absences from quizzes and exams will only be excused with written documentation from an appropriately accredited professional (e.g., a medical doctor).

UD ACADEMIC POLICIES

As a general rule, I will follow the University of Delaware's policies on any issues that are not spelled out in this syllabus.

THE SAFETY OF OUR LEARNING ENVIRONMENT

Student learning can only occur when students and their instructors feel safe, respected, and supported by each other. To ensure that our learning environment is as safe as possible, you are expected to abide by the most up-to-date University of Delaware's COVID-19 Guidelines,

ACADEMIC HONESTY

Please familiarize yourself with UD policies regarding academic dishonesty. The Academic Honesty Policy states: “Students must be honest and forthright in their academic studies. Students are expected to do their own work and must give proper credit for any work not their own. Students may neither give nor receive unauthorized assistance. Engaging in academic dishonesty, or allowing other students to do the same, corrupts the educational process and diminishes the quality of a University of Delaware degree.” View the university’s procedures for resolving academic dishonesty concerns. Contact Community Standards & Conflict Resolution at communitystandards@udel.edu or (302) 831-2117.

UNAUTHORIZED RECORDING AND INTELLECTUAL PROPERTY

In accordance with the Student Guide to University Policies, students are prohibited from (1) unauthorized recording, entirely or in part, of any class, and (2) posting of notes, study guides, exams, answer keys or any other materials from a class (whether the student is enrolled in the class or not) on the Internet, whether or not for a fee, without express permission from the faculty member. Any questions regarding the Student Guide to University Policies should be directed to the University’s Office of Community Standards & Conflict Resolution.

HARASSMENT, DISCRIMINATION, AND SEXUAL MISCONDUCT

The University of Delaware works to promote an academic and work environment that is free from all forms of discrimination, including harassment and sexual misconduct. As a member of the community, your rights, resources and responsibilities are reflected in the Non-Discrimination, Sexual Misconduct, and Title IX policy. Please familiarize yourself with this policy at the University’s Office of Equity & Inclusion’s website. You can report any concerns to the University’s Office of Equity & Inclusion (302) 831-8063 or at kbuchko@udel.edu. You can report anonymously through UD Police (302) 831-2222 or the EthicsPoint Compliance Hotline. (Read the full policy; File a report.)

RESPONSIBLE EMPLOYEE DISCLOSURE STATEMENT

As a University employee, I must report any alleged violation of the Non-Discrimination, Sexual Misconduct and Title IX Policy. This means that if I become aware of any instance in which a member of our campus community, including students, may have experienced sexual misconduct—including sexual harassment, sexual assault, relationship violence (domestic or dating violence), or stalking — or any form of discrimination as outlined in the Policy, I am required to report the matter to the University’s Title IX Coordinator. Once a report is made, the Title IX Coordinator will reach out to the affected party to provide information about their rights and options, available support resources, and how the University may respond to the

report. The goal is to ensure the safety and well-being of both the individual and the broader campus community.

If such information is shared with me — whether during class, in a written assignment, or in a one-on-one conversation — I will respect your privacy. I will share the information only with the Title IX Coordinator.

For more information about the University's Non-Discrimination, Sexual Misconduct and Title IX Policy, support services, and reporting procedures, please visit www.udel.edu/sexualmisconduct. You may also contact the Title IX Coordinator directly at kbuchko@udel.edu. The University offers the following confidential resources:

1. TimelyCare provides 24/7 mental health support via telehealth for confidential crisis and victim support for survivors of all forms of sexual misconduct. Students can receive eight (8) free scheduled telehealth counseling sessions per year with a therapist of their choosing. Students can register for TimelyCare at <https://www.udel.edu/students/health-wellbeing/counseling-center-services/timelycare/>
2. If you need immediate counseling assistance after hours, TimelyCare: TalkNow offers unlimited free emergency virtual sessions within five (5) minutes of logging on. Available 24/7, TalkNow provides on-demand emotional support to talk about anything, including anxiety, relationships, depression and school-related issues. TalkNow can be accessed at <https://www.udel.edu/students/health-wellbeing/counseling-center-services/timelycare/>
3. Student Wellness & Health Promotion offers one-on-one appointments with a professional victim advocate or a dating violence counselor. The Victim Advocate is available for all UD students facing traumatic situations, including sexual assault, dating and domestic violence, sexual harassment, and stalking. The Victim Advocate also provides information on resources including professional advocacy, support groups and 24/7 crisis assistance. To contact the Victim Advocate, email victimsupport@udel.edu, use this link to schedule an appointment: Book an Appointment with me or visit <https://www.udel.edu/students/support/victim-support-services/> for more information.
4. The Center for Counseling & Student Development Services provides individual and group counseling. The Counseling Center also offers walk-in hours Monday-Friday between 9:30am-4pm that are based on a first-come, first-served basis.

ACCOMMODATIONS FOR STUDENTS WITH DISABILITIES

Any student who may need disability-related accommodations should contact the Office of Disability Support Services (DSS) office as soon as possible. For more information, please visit www.udel.edu/dss. Contact DSS by phone: 302-831-4643; fax: 302-831-3261; website: www.udel.edu/dss; email: dssoffice@udel.edu; or visit 240 Academy Street, Alison Hall Suite 130 during business hours (8-5 M-F).

NOTICE OF NON-DISCRIMINATION AND EQUAL OPPORTUNITY

The University of Delaware does not discriminate against any person on the basis of race, color, national origin, sex, gender identity or expression, sexual orientation, genetic information, marital status, disability, religion, age, veteran status or any other characteristic protected by applicable law in its employment, educational programs and activities, admissions, and scholarship and loan programs as required by Title IX of the Educational Amendments of 1972, the Americans with Disabilities Act of 1990, Section 504 of the Rehabilitation Act of 1973, Title VI and VII of the Civil Rights Act of 1964, and other applicable statutes and University policies. The University of Delaware also prohibits unlawful harassment, including sexual harassment and sexual violence.

For inquiries or complaints related to non-discrimination policies, please contact: Office of Equity & Inclusion- oei@udel.edu, (302) 831-8063

For complaints related to Section 504 of the Rehabilitation Act of 1973 and/or the Americans with Disabilities Act, please contact: Office of Disability Support Services, dssoffice@udel.edu, Alison Hall, Suite 130, Newark, DE 19716 (302) 831-4643 OR contact the U.S. Department of Education - Office for Civil Rights.

ACKNOWLEDGMENTS

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